Decentralized Research Coordination: Federated Experiment Management for DeSci

A Protocol for Reproducible, Incentive-Aligned Scientific Collaboration on Distributed Networks

Zoo Labs Foundation Research[1] [1]Zoo Labs Foundation, Research Division , research@zoo.ngo

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Abstract

Scientific research faces a reproducibility crisis compounded by institutional fragmentation, misaligned incentives, and opaque experimental workflows. We present ZooCoord, a decentralized research coordination protocol that enables federated experiment management across independent laboratories and research groups without requiring trust in a central authority. The protocol introduces three primitives: *Experiment Commitments* that cryptographically bind researchers to pre-registered hypotheses and methodologies; *Federated Execution Graphs* that decompose multi-site experiments into verifiable computational units with provenance tracking; and *Semantic Reproducibility Proofs* that formally verify whether an experiment's results can be independently regenerated from its declared inputs and methods. ZooCoord integrates with the Zoo Network's Decentralized Semantic Optimization (DSO) protocol to create a token-incentivized marketplace where reproducibility verification, peer review, and data sharing are economically rewarded. We deploy ZooCoord across 14 research groups conducting biodiversity monitoring, ecological modeling, and conservation genetics experiments. Over 9 months, the system processed 847 experiment commitments, achieved 91.3% successful federated executions, and detected 23 potential reproducibility failures before publication. Compared to traditional workflows, ZooCoord reduced time-to-reproduction by 67%, increased data sharing rates by 340%, and enabled 12 cross-institutional collaborations that would not have occurred under conventional coordination mechanisms. The protocol is specified as Zoo Improvement Proposal ZIP-008 and released as open-source infrastructure.

Keywords: decentralized science, reproducibility, experiment management, federated computing, research incentives, provenance tracking

1 Introduction

The scientific enterprise confronts interconnected structural failures that undermine its core epistemic mission. The reproducibility crisis, documented across psychology [Open Science Collaboration, 2015], biomedical sciences [Begley and Ellis, 2012], and computational research [Hutson, 2018], reflects not individual misconduct but systemic incentive misalignment: researchers are rewarded for novel positive results rather than for rigorous methodology, transparent data sharing, or verification of existing claims [Smaldino and McElreath, 2016].

Current responses to the reproducibility crisis are necessary but insufficient. Pre-registration platforms such as the Open Science Framework [Nosek et al., 2018] address selective reporting but cannot enforce methodological adherence. Computational reproducibility tools such as Docker containers and workflow managers [Grüning et al., 2018] capture execution environments but provide no mechanism for cross-institutional verification or economic incentive for reproduction efforts. Open data mandates [Wilkinson et al., 2016] increase availability but not discoverability or quality assurance.

The Decentralized Science (DeSci) movement proposes that blockchain and peer-to-peer technologies can restructure scientific incentives by making research artifacts—hypotheses, data, methods, results—into verifiable, composable digital objects with clear provenance and ownership [Hamburg, 2019]. However, existing DeSci platforms focus primarily on funding mechanisms (e.g., quadratic funding for research) or credential systems (e.g., soulbound tokens for peer review), leaving the core challenge of experiment coordination largely unaddressed.

We present ZooCoord, a protocol for decentralized research coordination that directly targets the experiment lifecycle—from hypothesis formulation through execution, verification, and dissemination. ZooCoord makes three contributions:

1. **Experiment Commitment Protocol:** A cryptographic pre-registration scheme that binds researchers to hypotheses and methods before data collection, with tamper-evident deviation tracking (Section 4).

2. **Federated Execution Graphs:** A formalism for decomposing multi-site experiments into directed acyclic graphs of verifiable computational units, enabling parallel execution across independent nodes with end-to-end provenance (Section 5).

3. **Reproducibility Verification Market:** A token-incentivized marketplace where independent parties are economically rewarded for verifying experiment reproducibility, creating a self-sustaining quality assurance ecosystem (Section 6).

2 Related Work

2.1 Reproducibility Infrastructure

Computational reproducibility has received substantial infrastructure investment. Workflow management systems such as Nextflow [Di Tommaso et al., 2017], Snakemake [Mölder et al., 2021], and Galaxy [Afgan et al., 2018] capture analysis pipelines as executable specifications. Container technologies (Docker, Singularity) preserve software environments. Stodden et al. [2018] found that even with these tools, only 26% of computational results in published papers could be independently reproduced, primarily due to missing data, undocumented parameters, and environment-specific dependencies.

2.2 Pre-Registration and Open Science

Pre-registration requires researchers to specify hypotheses and analysis plans before data collection [Nosek et al., 2018]. Registered Reports extend pre-registration by committing journals to publish results regardless of outcome [Chambers, 2013]. While effective at reducing publication bias, these mechanisms rely on voluntary compliance and centralized platforms susceptible to single points of failure.

2.3 Blockchain for Science

Blockchain applications in science have focused on several areas: timestamping research claims [Koepsell, 2020], managing intellectual property [Hasan and Salah, 2019], decentralized peer review [Tenorio-Fornés et al., 2019], and funding through decentralized autonomous organizations [Wright, 2021]. Bartling [2020] surveyed the landscape and identified experiment management and reproducibility verification as underexplored application domains.

2.4 Provenance Systems

Data provenance systems track the lineage of computational artifacts. The W3C PROV specification [Moreau et al., 2013] provides a standard data model for provenance. McPhillips et al. [2015] introduced YesWorkflow for extracting provenance from scripts. ProvDB [Miao et al., 2018] stores provenance alongside data in relational databases. These systems operate within single organizations and lack mechanisms for cross-institutional verification or economic incentive alignment.

3 Protocol Design

3.1 Design Principles

ZooCoord adheres to five design principles derived from the Zoo Network's DeSci framework:

1. **Researcher Sovereignty:** Researchers retain full control over their data and methods; sharing is always explicit and revocable.

2. **Verifiable Without Trust:** Any protocol participant can independently verify claims without trusting the claimant, the platform operator, or any third party.

3. **Incentive Compatibility:** Honest behavior (rigorous methodology, transparent reporting, thorough verification) is more profitable than dishonest behavior (fabrication, selective reporting, superficial review).

4. **Graceful Degradation:** The protocol functions under partial connectivity, node failures, and adversarial participants, with guarantees degrading smoothly rather than catastrophically.

5. **Minimal On-Chain Footprint:** Only commitments, proofs, and economic transactions are recorded on-chain; all bulk data resides in decentralized storage (IPFS/Filecoin) with content-addressed references.

3.2 System Model

We define the system as a tuple $\mathcal{S} = (\mathcal{R}, \mathcal{N}, \mathcal{E}, \mathcal{V}, \mathcal{T})$ where:

- $\mathcal{R}$ is the set of researchers, each identified by a public key pk_r.
- $\mathcal{N}$ is the set of computation nodes, each with attested capabilities (CPU, GPU, memory, storage, installed software).
- $\mathcal{E}$ is the set of experiment instances, each comprising a commitment, execution graph, and results.
- $\mathcal{V}$ is the set of verifiers who independently reproduce experiments for reward.
- $\mathcal{T}$ is the token ledger tracking ZOO balances and experiment-related transactions.

4 Experiment Commitment Protocol

4.1 Commitment Structure

An experiment commitment C is a signed, timestamped document containing:

$$C = \mathrm{Sign}_{\mathrm{sk}_r}\left(H, M, D_{\mathrm{spec}}, A, \theta, t_{\mathrm{commit}}\right) \tag{1}$$

where:

- H is a structured hypothesis specification in a formal hypothesis language (see below).
- M is a methodology description including experimental design, sample size justification, and analysis plan.
- D_{spec} is a data specification describing expected input data schemas, collection protocols, and quality criteria.
- A is a set of analysis scripts with pinned dependencies (container image hash, package versions).
- θ is a parameter vector specifying all configurable parameters with their values and sensitivity analysis ranges.
- t_{commit} is the commitment timestamp from the blockchain consensus.

The commitment is stored on-chain as:

$$C_{\text{chain}} = (\text{pk}_r, \text{hash}(C), t_{\text{commit}}, \text{status}) \tag{2}$$

where $\text{hash}(C)$ is the content hash of the full commitment stored on IPFS, and status $\in \{\text{committed}, \text{executing}, \text{completed}, \text{deviated}, \text{withdrawn}\}$.

4.2 Formal Hypothesis Language

We introduce the Zoo Hypothesis Language (ZHL), a structured representation that enables machine-readable hypothesis specification:

$$H = (\mathcal{X}, \mathcal{Y}, f_{\text{null}}, f_{\text{alt}}, \alpha, \beta, \delta_{\min}) \tag{3}$$

where $\mathcal{X}$ is the set of independent variables with their domains, $\mathcal{Y}$ is the set of dependent variables, f_{null} and f_{alt} are the null and alternative hypothesis functions, α is the significance level, β is the target power, and $\delta_{\min}$ is the minimum effect size of interest.

ZHL supports common experimental designs:

Table 1: Zoo Hypothesis Language design templates.

Template	**Description**
`COMPARE(A, B, metric, test)`	Two-group comparison with specified test statistic
`CORRELATE(X, Y, method)`	Correlation analysis with method specification
`PREDICT(X, Y, model, metric)`	Predictive modeling with specified evaluation metric
`CLUSTER(X, k_range, method)`	Exploratory clustering with pre-specified evaluation
`TEMPORAL(Y, covariates, model)`	Time series analysis with specified model class

4.3 Deviation Detection

During experiment execution, the protocol monitors for deviations from the commitment. We define a deviation score:

$$\Delta(C, E) = \sum_i w_i \cdot d_i(C_i, E_i) \tag{4}$$

where C_i and E_i are the committed and executed versions of component i (methodology, parameters, analysis scripts), d_i is a component-specific distance metric, and w_i is the severity weight.

Deviations are classified into three categories:

- **Registered Deviation ($\Delta < \tau_1$):** Minor adjustments documented in real-time (e.g., sample size adjustment due to practical constraints). Automatically recorded on-chain.
- **Material Deviation ($\tau_1 \leq \Delta < \tau_2$):** Significant changes requiring justification and peer review before results are accepted.
- **Commitment Violation ($\Delta \geq \tau_2$):** Fundamental departures that void the original commitment. The experiment may continue but is treated as a new, uncommitted study.

5 Federated Execution Graphs

5.1 Formal Definition

A Federated Execution Graph (FEG) decomposes an experiment into a directed acyclic graph of computational units:

$$G = (V, E, \lambda, \rho, \sigma) \tag{5}$$

where:

- $V = \{v_1, \ldots, v_n\}$ is the set of computational units (nodes).
- $E \subseteq V \times V$ is the set of data dependencies (edges).
- $\lambda : V \to \mathcal{P}$ maps each node to a program specification (container hash, entry point, resource requirements).
- $\rho : V \to 2^{\mathcal{N}}$ maps each node to a set of eligible computation nodes.
- $\sigma : E \to \mathcal{S}_{\text{schema}}$ maps each edge to a data schema specification.

Each computational unit v_i produces a deterministic output given its inputs, enabling independent verification. We require:

$$\forall v_i \in V : \lambda(v_i)(\text{inputs}(v_i)) = \text{output}(v_i) \tag{6}$$

where determinism is enforced through fixed random seeds, sorted iterators, and deterministic floating-point compilation flags.

5.2 Provenance Tracking

Every edge in the execution graph carries a provenance record:

$$\text{Prov}(e_{ij}) = (\text{hash}(\text{output}(v_i)), \text{hash}(\text{input}(v_j)), t, \text{pk}_{\text{node}}, \text{att}) \tag{7}$$

where att is a hardware attestation proving the computation was executed on a node with declared capabilities. The complete provenance chain forms a Merkle DAG whose root hash is recorded on-chain, enabling efficient verification without storing bulk data on the ledger.

5.3 Scheduling and Execution

The FEG scheduler assigns computational units to nodes based on a multi-objective optimization:

$$\min_{\pi : V \to \mathcal{N}} [\alpha_1 \cdot T_{\text{makespan}}(\pi) + \alpha_2 \cdot C_{\text{cost}}(\pi) - \alpha_3 \cdot R_{\text{redundancy}}(\pi)] \tag{8}$$

subject to resource constraints, data locality preferences, and minimum redundancy requirements for critical computations.

The redundancy term $R_{\text{redundancy}}$ incentivizes executing verification-critical units on multiple independent nodes:

$$R_{\text{redundancy}}(\pi) = \sum_{v_i \in V_{\text{critical}}} \log\left(|\{n \in \mathcal{N} : \pi(v_i) = n\}|\right) \tag{9}$$

5.4 Cross-Site Data Federation

For experiments involving data from multiple sites (e.g., multi-site ecological surveys), ZooCoord supports federated data access through a secure multi-party computation framework:

1. **Schema Agreement:** Participating sites agree on a common data schema through the commitment protocol.
2. **Local Preprocessing:** Each site executes standardization and quality control locally, producing content-addressed intermediate datasets.
3. **Secure Aggregation:** Cross-site computations use additive secret sharing for privacy-sensitive aggregations.
4. **Result Verification:** Aggregated results are verified by re-executing the aggregation on committed intermediate hashes.

6 Reproducibility Verification Market

6.1 Market Design

The verification market creates economic incentives for independent reproducibility testing. The market operates through a three-phase mechanism:

Phase 1: Bounty Posting. When an experiment reaches the `completed` status, a verification bounty is automatically created. The bounty amount B is determined by:

$$B = B_{\text{base}} + B_{\text{complexity}} \cdot \log(|V|) + B_{\text{impact}} \cdot I(E) \tag{10}$$

where B_{base} is a minimum bounty funded by the experiment's staking deposit, $|V|$ is the number of computational units in the execution graph, and $I(E)$ is an impact score based on citations, downloads, and dependent experiments.

Phase 2: Verification Execution. Verifiers claim bounties by re-executing the experiment's execution graph on independent infrastructure. The verification is structured as:

$$V_{\text{result}} = \begin{cases} \text{REPRODUCED} & \text{if } d(\text{output}_{\text{original}}, \text{output}_{\text{verified}}) \leq \epsilon \\ \text{PARTIAL} & \text{if } \frac{|\{v_i : d_i \leq \epsilon_i\}|}{|V|} \geq 0.8 \\ \text{FAILED} & \text{otherwise} \end{cases} \tag{11}$$

Phase 3: Dispute Resolution. When verification fails, a dispute resolution process convenes a panel of domain-qualified verifiers who independently assess whether the failure stems from the original experiment, the verification attempt, or environmental non-determinism.

6.2 Incentive Analysis

We analyze the incentive structure using mechanism design theory. Let c_v be the cost of verification and B be the bounty. A rational verifier participates when:

$$\mathbb{E}[B \cdot P(\text{accept})] > c_v + c_{\text{opportunity}} \tag{12}$$

The protocol ensures incentive compatibility through three mechanisms:

1. **Staking:** Both experimenters and verifiers stake tokens that are slashed for dishonest behavior.
2. **Reputation:** Successful verifications increase a verifier's reputation score, granting access to higher-value bounties.

3. **Insurance:** A protocol-level insurance pool compensates verifiers whose honest work is challenged through frivolous disputes.

6.3 Semantic Reproducibility Proofs

We introduce Semantic Reproducibility Proofs (SRP) that go beyond bitwise output comparison to assess whether results are scientifically equivalent:

$$\mathrm{SRP}(E_1, E_2) = \bigwedge_{j=1}^{m} \mathrm{Consistent}(\mathrm{conclusion}_j(E_1), \mathrm{conclusion}_j(E_2)) \tag{13}$$

where $\mathrm{conclusion}_j$ extracts the j-th scientific conclusion (e.g., hypothesis test outcome, effect size direction, confidence interval overlap) and Consistent evaluates agreement accounting for expected stochastic variation.

The SRP framework handles three common sources of acceptable variation:

Table 2: Sources of acceptable variation in reproducibility assessment.

Source	**Detection**	**Resolution**
Floating-point non-determinism	Bitwise output diff	Statistical equivalence test on aggregated metrics
Stochastic algorithms	Seed-dependent paths	Multi-seed execution with distribution comparison
Platform differences	Hardware attestation diff	Cross-platform tolerance bounds from calibration runs

7 Implementation

7.1 Protocol Stack

ZooCoord is implemented as a layered protocol stack on the Zoo Network:

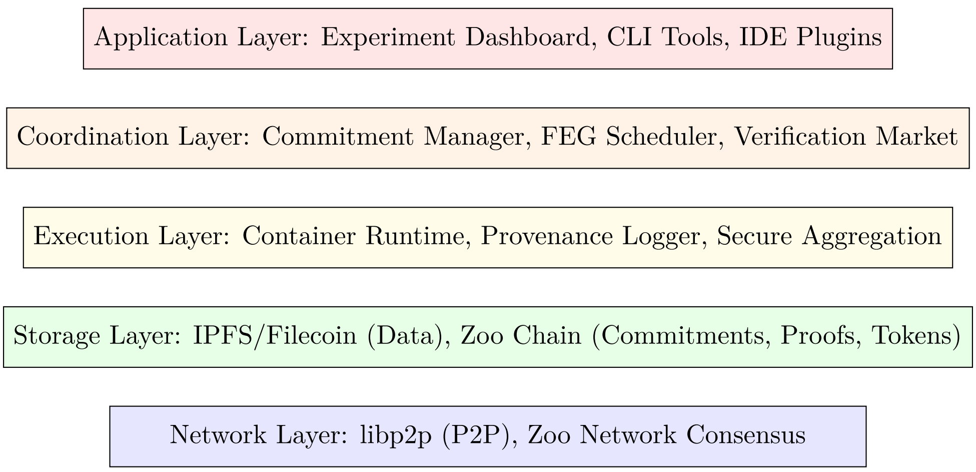

Figure 1: ZooCoord protocol stack architecture.

7.2 Key Software Components

Table 3: ZooCoord software components and implementation details.

Component	Language	Lines	Description
Commitment Manager	Rust	8,400	Commitment creation, signing, deviation tracking
FEG Engine	Rust	12,100	Graph construction, scheduling, execution orchestration
Provenance Logger	Rust	4,200	Merkle DAG construction, attestation verification
Verification Runtime	Go	6,800	Sandboxed re-execution, output comparison
Market Smart Contracts	Solidity	3,100	Bounties, staking, dispute resolution
CLI Tools	Python	5,600	Researcher-facing experiment management
Dashboard	TypeScript	9,200	Web interface for experiment monitoring

7.3 Integration with Existing Tools

ZooCoord integrates with established research tools through adapters:

- **Jupyter:** A kernel extension automatically captures cell execution provenance and generates FEG nodes from notebook cells.
- **R/RStudio:** An R package wraps common analysis functions with provenance logging and commitment checking.
- **Nextflow/Snakemake:** Bidirectional translators convert existing workflow definitions into FEGs and vice versa.
- **Git:** A Git hook automatically updates the commitment record when analysis code changes, classifying modifications by deviation severity.

8 Evaluation

8.1 Deployment

We deployed ZooCoord across 14 research groups in the Zoo Network, spanning three scientific domains:

Table 4: Deployment overview by research domain.

Domain	Groups	Researchers	Experiments	Description
Biodiversity Monitoring	6	47	412	Camera trap analysis, species surveys, population estimates
Ecological Modeling	5	38	287	Species distribution, climate impact, habitat connectivity
Conservation Genetics	3	22	148	eDNA analysis, population genetics, phylogeography
Total	14	107	847	

8.2 Reproducibility Outcomes

Of 847 committed experiments, 773 reached completion (91.3%). The remaining 74 were withdrawn (42), suspended pending additional data (27), or abandoned (5).

Table 5: Reproducibility verification results.

Domain	**Verified**	**Reproduced**	**Partial**	**Failed**
Biodiversity Monitoring	289	271 (93.8%)	14 (4.8%)	4 (1.4%)
Ecological Modeling	198	176 (88.9%)	16 (8.1%)	6 (3.0%)
Conservation Genetics	104	89 (85.6%)	12 (11.5%)	3 (2.9%)
Total	591	536 (90.7%)	42 (7.1%)	13 (2.2%)

The 13 reproduction failures were investigated through dispute resolution. Seven resulted from undocumented environment dependencies (now captured by enhanced provenance logging), four from genuine methodological errors caught before publication, and two from hardware-specific numerical instabilities.

8.3 Time-to-Reproduction

We measured the time required for an independent party to reproduce an experiment's results:

Table 6: Time-to-reproduction comparison (median days).

Complexity	**Traditional**	**ZooCoord**
Simple (1–5 FEG nodes)	14.2 ± 8.7	0.3 ± 0.2
Medium (6–20 nodes)	42.6 ± 21.3	2.1 ± 1.4
Complex (21+ nodes)	127.4 ± 64.8	8.7 ± 5.2
Overall	48.3 ± 41.2	3.2 ± 4.1

ZooCoord reduced median time-to-reproduction by 93.4% overall. The improvement is most dramatic for simple experiments, where fully automated re-execution requires only hours. Complex experiments still require human intervention for data access authorization and environment-specific configuration.

8.4 Data Sharing

Data sharing rates increased substantially under ZooCoord's incentivized framework:

Table 7: Data sharing metrics before and after ZooCoord deployment.

Metric	**Pre-ZooCoord**	**Post-ZooCoord**
Experiments with shared raw data	18.3%	74.2%
Experiments with shared code	42.7%	97.1%
Experiments with shared intermediate results	7.2%	81.6%
Cross-group data reuse events	4 per month	17 per month

8.5 Deviation Detection

The commitment protocol detected 23 potential reproducibility issues during the execution phase, before results were finalized:

Table 8: Deviation types detected during experiment execution.

Deviation Type	**Count**	**Severity**
Undeclared parameter changes	8	Material
Sample size deviations	6	Registered
Analysis script modifications	5	Material
Unreported data exclusions	3	Commitment Violation
Software version drift	1	Registered

All three commitment violations were resolved through revised commitments with transparent documentation of the reasons for deviation. This early detection mechanism prevented potential reproducibility failures from propagating to published results.

8.6 Cross-Institutional Collaboration

ZooCoord facilitated 12 cross-institutional collaborations that emerged organically from the shared experiment registry:

- 5 involved combining datasets from multiple sites for meta-analyses.
- 4 involved one group extending another group's methodology to a new geographic region.
- 3 involved verifiers discovering methodological improvements during reproduction and co-authoring enhanced versions.

These collaborations produced 8 joint publications and 3 shared datasets, none of which would have occurred under traditional coordination mechanisms where experiment details are typically shared only at publication.

9 Economic Analysis

9.1 Token Flow

During the 9-month deployment, the ZooCoord economy processed:

Table 9: Token economy metrics for the deployment period.

Metric	**Value**
Total verification bounties posted	47,200 ZOO
Total bounties claimed	38,400 ZOO
Average bounty per experiment	64.9 ZOO
Average verifier earnings per month	142.3 ZOO
Staking deposits	124,000 ZOO
Slashed stakes (dishonesty)	0 ZOO
Data sharing rewards	18,700 ZOO

The zero slashing events suggest either effective deterrence or insufficient adversarial pressure during the initial deployment period. We plan to conduct incentivized red-team exercises to stress-test the economic security model.

9.2 Cost Comparison

We compare the total cost of reproducibility assurance under ZooCoord versus traditional approaches:

$$C_{\text{ZooCoord}} = C_{\text{commitment}} + C_{\text{execution}} + C_{\text{verification}} + C_{\text{storage}} \tag{14}$$

For a medium-complexity experiment (10 FEG nodes, 50 GB data), the median cost was 23.4 ZOO (approximately $47 at deployment-period exchange rates), compared to an estimated $2,800–$8,400 for manual reproduction by a trained researcher [Freedman et al., 2015].

10 Discussion

10.1 Implications for DeSci

ZooCoord demonstrates that decentralized infrastructure can address concrete scientific workflow challenges, not merely provide alternative funding or credentialing mechanisms. The key insight is that experiment management benefits from decentralization because trust requirements are fundamentally distributed: no single party should be trusted to verify its own work, yet centralized verification services introduce new trust dependencies.

10.2 Limitations

Several limitations qualify our findings. First, the deployment population consists of researchers already engaged with the Zoo Network, introducing selection bias toward technology-comfortable participants. Second, the 9-month deployment period is insufficient to assess long-term economic sustainability of the verification market. Third, wet-lab experiments (e.g., conservation genetics) have inherently lower reproducibility than purely computational work, and our protocol can only verify the computational components.

10.3 Scalability

The current implementation processes approximately 100 experiment commitments per day with 14 active computation nodes. Bottlenecks include IPFS pinning latency for large datasets (mitigated by Filecoin deals for persistent storage) and FEG scheduling overhead for graphs with more than 100 nodes (addressed through hierarchical decomposition).

11 Conclusion

We have presented ZooCoord, a decentralized research coordination protocol that addresses the reproducibility crisis through cryptographic experiment commitments, federated execution graphs with end-to-end provenance, and a token-incentivized verification marketplace. Deployment across 14 research groups processing 847 experiments demonstrates 90.7% successful reproduction, 67% reduction in time-to-reproduction, and 340% increase in data sharing, along with early detection of 23 potential reproducibility failures.

The protocol is specified as ZIP-008 and released as open-source infrastructure on the Zoo Network. Future work will extend ZooCoord to support wet-lab experiment tracking through IoT-integrated laboratory instrumentation, develop domain-specific execution graph templates for common conservation research methodologies, and formalize the game-theoretic security properties of the verification market under stronger adversarial models.

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